

Multi-parameter fractional integration on Heisenberg group

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Abstract

We study a family of fractional integral operator defined as

$$\mathbf{I}_{\alpha\beta\vartheta}f(u, v, t) = \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \mathbf{V}^{\alpha\beta\vartheta}[(u, v, t) \odot (\xi, \eta, \tau)^{-1}] d\xi d\eta d\tau$$

where $\odot$ denotes the multiplication law.

$\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ satisfying Zygmund dilations. A characterization is established between $\mathbf{I}_{\alpha\beta\vartheta}: \mathbf{L}^p(\mathbb{R}^{2n+1}) \longrightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ and necessary constraints consisting of $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq 0$ for $1 < p < q < \infty$.

For $0 \leq \gamma < 1$, define

$$\mathbf{M}_\gamma f(u, v, t) = \sup_{\mathbf{R} \subset \mathbb{R}^{2n+1}} \text{vol}(\mathbf{R})^{\gamma-1} \iint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau.$$

where $\mathbf{R} \subset \mathbb{R}^{2n+1}$ is a rectangle centered on the origin with sides parallel to the coordinates. We show $\mathbf{M}_\gamma: \mathbf{L}^p(\mathbb{R}^{2n+1}) \longrightarrow \mathbf{L}^q(\mathbb{R}^{2n+1})$ for $1 < p \leq q < \infty$ if and only if $\gamma = \frac{1}{p} - \frac{1}{q}$.

1 Introduction

A fractional integral operator $\mathbf{T}_a$ is initially defined on $\mathbb{R}^N$ as

$$\mathbf{T}_a f(x) = \int_{\mathbb{R}^N} f(y) \left[\frac{1}{|x-y|} \right]^{N-a} dy, \quad 0 < a < N. \quad (1.1)$$

In 1928, Hardy and Littlewood [1] have obtained an regularity theorem for $\mathbf{T}_a$ when $N = 1$. Ten years later, Sobolev [2] made extensions on every higher dimensional space.

◊ Throughout, $\mathfrak{B} > 0$ is regarded as a generic constant depending on its sub-indices.

Hardy-Littlewood-Sobolev theorem *Let $\mathbf{T}_a$ defined in (1.1) for $0 < a < N$. We have*

$$\begin{aligned} \|\mathbf{T}_a f\|_{\mathbf{L}^q(\mathbb{R}^N)} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^N)}, \quad 1 < p < q < \infty \\ \text{if and only if} \quad \frac{a}{N} &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1.2)$$

This classical result was first re-investigated by Folland and Stein [4] on Heisenberg group. We shall work on its real variable representation with a multiplication law:

$$\begin{aligned} (u, v, t) \odot (\xi, \eta, \tau) &= \left[u + \xi, v + \eta, t + \tau + \mu(u \cdot \eta - v \cdot \xi) \right], \quad \mu \in \mathbb{R} \\ (u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}, \quad (\xi, \eta, \tau)^{-1} &= (-\xi, -\eta, -\tau) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}. \end{aligned} \quad (1.3)$$

Let $0 < \delta < n + 1$. Consider

$$\mathbf{S}_\delta f(u, v, t) = \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \Omega^\delta[(u, v, t) \odot (\xi, \eta, \tau)^{-1}] d\xi d\eta d\tau. \quad (1.4)$$

Ω^δ is a distribution in $\mathbb{R}^{2n+1}$ agree with

$$\Omega^\delta(u, v, t) = \left[\frac{1}{|u|^2 + |v|^2 + |t|} \right]^{n+1-\delta}, \quad (u, v, t) \neq (0, 0, 0). \quad (1.5)$$

Folland-Stein theorem Let $\mathbf{S}_\delta$ defined in (1.4)-(1.5) for $0 < \delta < n + 1$. We have

$$\begin{aligned} \|\mathbf{S}_\delta f\|_{\mathbf{L}^q(\mathbb{R}^{n+1})} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p < q < \infty \\ \text{if and only if} \quad \frac{\delta}{n+1} &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1.6)$$

The best constant for the $\mathbf{L}^p \rightarrow \mathbf{L}^q$ -norm inequality in (1.6) is found by Frank and Lieb [13]. A discrete analogue of this result has been obtained by Pierce [14]. Recently, the regarding commutator estimates are established by Fanelli and Roncal [15].

In this paper, we give a multi-parameter extension to **Folland-Stein theorem** by replacing Ω^δ with a larger kernel having singularity on every coordinate subspace. From (1.5), we have

$$\Omega^\delta(u, v, t) \leq \left[\frac{1}{|u||v| + |t|} \right]^{n+1-\delta}, \quad (u, t) \neq (0, 0) \text{ or } (v, t) \neq (0, 0).$$

A direct computation shows

$$\begin{aligned} \left[\frac{1}{|u||v| + |t|} \right]^{n+1-\delta} &\approx \left[\frac{1}{|u|^2|v|^2 + t^2} \right]^{\frac{n+1}{2}-\frac{\delta}{2}} = |u|^{\frac{\delta}{2}-\frac{n+1}{2}} |v|^{\frac{\delta}{2}-\frac{n+1}{2}} |t|^{\frac{\delta}{2}-\frac{n+1}{2}} \left[\frac{|u||v||t|}{|u|^2|v|^2 + t^2} \right]^{\frac{n+1}{2}-\frac{\delta}{2}} \\ &= |u|^{\left[\frac{\delta}{2}+\frac{n-1}{2}\right]-n} |v|^{\left[\frac{\delta}{2}+\frac{n-1}{2}\right]-n} |t|^{\left[\frac{\delta}{2}-\frac{n-1}{2}\right]-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\left[\frac{n+1}{2}-\frac{\delta}{2}\right]}, \\ &\quad u \neq 0, v \neq 0, t \neq 0. \end{aligned}$$

Above estimates lead us to the following assertion. Let $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq 0$. $\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ agree with

$$\mathbf{V}^{\alpha\beta\vartheta}(u, v, t) = |u|^{\alpha-n} |v|^{\alpha-n} |t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\vartheta}, \quad u \neq 0, v \neq 0, t \neq 0. \quad (1.7)$$

Remark 1.1. By taking into account $\alpha = \frac{\delta}{2} + \frac{n-1}{2}$, $\beta = \frac{\delta}{2} - \frac{n-1}{2}$ and $\vartheta = \frac{n+1}{2} - \frac{\delta}{2}$ for $0 < \delta < n + 1$, we find $\alpha > n\beta$ and $\vartheta = \frac{n+1}{2} - \frac{\delta}{2} > \frac{\alpha-n\beta}{n+1}$.

Define

$$\mathbf{I}_{\alpha\beta\vartheta} f(u, v, t) = \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau) \mathbf{V}^{\alpha\beta\vartheta}[(u, v, t) \odot (\xi, \eta, \tau)^{-1}] d\xi d\eta d\tau. \quad (1.8)$$

Observe that

$$\mathbf{V}^{\alpha\beta\vartheta}[(ru, sv, rst) \odot (r\xi, s\eta, r\tau)^{-1}] = r^{\alpha+\beta-n-1} s^{\alpha+\beta-n-1} \mathbf{V}^{\alpha\beta\vartheta}[(u, v, t) \odot (\xi, \eta, \tau)^{-1}], \quad r, s > 0. \quad (1.9)$$

A convolution operator of this type is said to be associated with Zygmund dilation. Singular integral operators carrying certain multi-parameter structures defined on Heisenberg group have been systematically studied, for instance by Phong and Stein [5], Ricci and Stein [6] and Müller, Ricci and Stein [7]. Much less is known for fractional integration in this direction.

Theorem One *Let $\mathbf{I}_{\alpha\beta\vartheta}$ defined in (1. 7)-(1. 8) for $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq 0$. We have*

$$\begin{aligned} \|\mathbf{I}_{\alpha\beta\vartheta} f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p < q < \infty \\ \text{if and only if} \quad \vartheta &\geq \frac{|\alpha - n\beta|}{n+1}, \quad \frac{\alpha + \beta}{n+1} = \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1. 10)$$

Next, denote $\mathbf{R} \subset \mathbb{R}^{2n+1}$ to be a rectangle parallel to the coordinates. Let $0 \leq \gamma < 1$. A strong fractional maximal operator is defined on Heisenberg group as

$$\mathbf{M}_\gamma f(u, v, t) = \sup_{\mathbf{R} \ni (0,0,0)} \text{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau. \quad (1. 11)$$

Theorem Two *Let $\mathbf{M}_\gamma$ defined in (1. 11) for $0 \leq \gamma < 1$. We have*

$$\begin{aligned} \|\mathbf{M}_\gamma f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p \leq q < \infty \\ \text{if and only if} \quad \gamma &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1. 12)$$

For $\gamma = 0$, $\mathbf{M}_0 \doteq \mathbf{M}$ in (1. 11) is the strong maximal operator defined on Heisenberg group. The $\mathbf{L}^p$ -boundedness of $\mathbf{M}$ is proved by Christ [11]. Thereby, the elegant work is done by using a mixture of techniques developed previously by Ricci and Stein [8]-[9] and Christ [12] for singular integrals defined on sub-manifolds within a general setting of Nilpotent Lie groups. We prove **Theorem Two** with a more direct approach by applying a multi-parameter covering lemma due to Córdoba and Fefferman [3].

As a special case, consider $\mathbf{R} = \mathbf{Q}_1 \times \mathbf{Q}_2 \times \mathbf{Q}_3 \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$: $\mathbf{Q}_1, \mathbf{Q}_2$ and $\mathbf{Q}_3$ are cubes centered on the origin of regarding subspaces. For $\alpha, \beta \in \mathbb{R}$, we define

$$\begin{aligned} \mathbf{M}_{\alpha\beta} f(u, v, t) &= \sup_{\mathbf{R} \ni (0,0,0): \text{vol}\{\mathbf{Q}_3\} = \text{vol}\{\mathbf{Q}_1\}^{\frac{1}{n}} \text{vol}\{\mathbf{Q}_2\}^{\frac{1}{n}}} \text{vol}\{\mathbf{Q}_1\}^{\frac{\alpha}{n}-1} \text{vol}\{\mathbf{Q}_2\}^{\frac{\beta}{n}-1} \text{vol}\{\mathbf{Q}_3\}^{\beta-1} \\ &\quad \iiint_{\mathbf{R}} |f[(u, v, t) \odot (\xi, \eta, \tau)^{-1}]| d\xi d\eta d\tau. \end{aligned} \quad (1. 13)$$

This is known as the fractional maximal function associated with Zygmund dilation defined on Heisenberg group. For $\mathbf{M}_{\alpha\beta}$ defined in Euclidean space with $\alpha = \beta = 0$, the regarding $\mathbf{L}^p$ -regularity theorem and its weighted analogue have been well established. See the papers by Ricci and Stein [6] and Fefferman and Pipher [10].

Later, we shall find

$$\mathbf{M}_{\alpha\beta} f(u, v, t) \leq \mathbf{M}_\gamma f(u, v, t), \quad \gamma = \frac{\alpha+\beta}{n+1}. \quad (1. 14)$$

Corollary One Let $\mathbf{M}_{\alpha\beta}$ defined in (1. 13) for $\alpha, \beta \in \mathbb{R}$. We have

$$\begin{aligned} \|\mathbf{M}_{\alpha\beta} f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} &\leq \mathfrak{B}_{p,q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}, \quad 1 < p \leq q < \infty \\ \text{if and only if} \quad \frac{\alpha + \beta}{n + 1} &= \frac{1}{p} - \frac{1}{q}. \end{aligned} \quad (1. 15)$$

The remaining paper is organized as follows. In the next section, we show that those two constraints consisting of $\alpha, \beta, \vartheta, p, q$ in (1. 10) are necessary conditions. In section 3, we prove **Theorem One**. In section 4, we prove **Theorem Two**.

2 Some necessary constraints

Let $\mathbf{I}_{\alpha\beta\vartheta}$ defined in (1. 7)-(1. 8). By changing variable $\tau \longrightarrow \tau + \mu(u \cdot \eta - v \cdot \xi)$, we find

$$\begin{aligned} \mathbf{I}_{\alpha\beta\vartheta} f(u, v, t) &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \mathbf{V}^{\alpha\beta\vartheta}(u - \xi, v - \eta, t - \tau) d\xi d\eta d\tau \\ &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \\ &\quad |u - \xi|^{\alpha-n} |v - \eta|^{\alpha-n} |t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau. \end{aligned} \quad (2. 1)$$

Consider a more general situation by replacing $\mathbf{V}^{\alpha\beta\vartheta}(u, v, t)$ with

$$|u|^{\alpha_1-n} |v|^{\alpha_2-n} |t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\vartheta}, \quad \alpha_1, \alpha_2, \beta \in \mathbb{R}, \quad \vartheta \geq 0. \quad (2. 2)$$

By changing dilations $(u, v, t) \longrightarrow (ru, sv, rs\lambda t)$ and $(\xi, \eta, \tau) \longrightarrow (r\xi, s\eta, rs\lambda\tau)$ for $r, s > 0$ and $0 < \lambda < 1$ or $\lambda > 1$, we have

$$\begin{aligned} &\left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iiint_{\mathbb{R}^{2n+1}} f \left[r^{-1}\xi, s^{-1}\eta, r^{-1}s^{-1}\lambda^{-1}[\tau - \mu\lambda(u \cdot \eta - v \cdot \xi)] \right] \right. \right. \\ &\quad \left. \left. |u - \xi|^{\alpha_1-n} |v - \eta|^{\alpha_2-n} |t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \right\}^q dudvdt \right\}^{\frac{1}{q}} \\ &= r^{\alpha_1+\beta} s^{\alpha_2+\beta} r^{\frac{n+1}{q}} s^{\frac{n+1}{q}} \lambda^{\beta} \lambda^{\frac{1}{q}} \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \right. \right. \\ &\quad \left. \left. |u - \xi|^{\alpha_1-n} |v - \eta|^{\alpha_2-n} |t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{\lambda|t - \tau|} + \frac{\lambda|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \right\}^q dudvdt \right\}^{\frac{1}{q}} \quad (2. 3) \\ &\geq r^{\alpha_1+\beta} s^{\alpha_2+\beta} r^{\frac{n+1}{q}} s^{\frac{n+1}{q}} \lambda^{\beta} \lambda^{\frac{1}{q}} \begin{cases} \lambda^{\vartheta}, & 0 < \lambda < 1, \\ \lambda^{-\vartheta}, & \lambda > 1 \end{cases} \\ &\quad \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \right. \right. \\ &\quad \left. \left. |u - \xi|^{\alpha_1-n} |v - \eta|^{\alpha_2-n} |t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \right\}^q dudvdt \right\}^{\frac{1}{q}}. \end{aligned}$$

The $\mathbf{L}^p \longrightarrow \mathbf{L}^q$ -norm inequality in (1. 12) implies that the last line of (2. 3) is bounded by

$$\left\{ \iiint_{\mathbb{R}^{2n+1}} \left| f\left(r^{-1}\xi, s^{-1}\eta, r^{-1}s^{-1}\lambda^{-1}\tau\right) \right|^p d\xi d\eta d\tau \right\}^{\frac{1}{p}} = r^{\frac{n+1}{p}} s^{\frac{n+1}{p}} \lambda^{\frac{1}{p}} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \quad (2. 4)$$

This must be true for every $r, s > 0$ and $0 < \lambda < 1$ or $\lambda > 1$. We necessarily have

$$\frac{\alpha_1 + \beta}{n+1} = \frac{1}{p} - \frac{1}{q} = \frac{\alpha_2 + \beta}{n+1}, \quad \beta + \vartheta \geq \frac{1}{p} - \frac{1}{q} \quad \text{or} \quad \beta - \vartheta \leq \frac{1}{p} - \frac{1}{q}. \quad (2. 5)$$

The first constraint in (2. 5) forces us to have $\alpha_1 = \alpha_2$. Therefore, write

$$\frac{\alpha + \beta}{n+1} = \frac{1}{p} - \frac{1}{q}, \quad \alpha = \alpha_1 = \alpha_2. \quad (2. 6)$$

By bringing (2. 6) to the two inequalities in (2. 5), we find

$$\vartheta \geq \beta - \frac{\alpha + \beta}{n+1} = \frac{n\beta - \alpha}{n+1} \quad \text{or} \quad \vartheta \geq \frac{\alpha + \beta}{n+1} - \beta = \frac{\alpha - n\beta}{n+1}. \quad (2. 7)$$

Together, we conclude $\vartheta \geq \frac{|\alpha - n\beta|}{n+1}$.

3 Proof of Theorem One

Given $\alpha, \beta \in \mathbb{R}$ and $\vartheta \geq \frac{|\alpha - n\beta|}{n+1}$, $\mathbf{V}^{\alpha\beta\vartheta}$ is a distribution in $\mathbb{R}^{2n+1}$ agree with $\mathbf{V}^{\alpha\beta\vartheta}(u, v, t)$ in (1. 7) whenever $u \neq 0, v \neq 0, t \neq 0$.

Suppose $\alpha \geq n\beta$. We have $\frac{|\alpha - n\beta|}{n+1} = \frac{\alpha - n\beta}{n+1}$ and

$$\begin{aligned} \mathbf{V}^{\alpha\beta\vartheta}(\xi, \eta, \tau) &\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{-\frac{\alpha-n\beta}{n+1}} \\ &\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|u||v|}{|t|} \right]^{-\frac{\alpha-n\beta}{n+1}} \\ &= |u|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t|^{\frac{\alpha+\beta}{n+1}-1}, \quad u \neq 0, v \neq 0, t \neq 0. \end{aligned} \quad (3. 1)$$

Suppose $\alpha \leq n\beta$. We find $\frac{|\alpha - n\beta|}{n+1} = \frac{n\beta - \alpha}{n+1}$ and

$$\begin{aligned} \mathbf{V}^{\alpha\beta\vartheta}(u, v, t) &\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|u||v|}{|t|} + \frac{|t|}{|u||v|} \right]^{\frac{\alpha-n\beta}{n+1}} \\ &\leq |u|^{\alpha-n}|v|^{\alpha-n}|t|^{\beta-1} \left[\frac{|t|}{|u||v|} \right]^{\frac{\alpha-n\beta}{n+1}} \\ &= |u|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|v|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n}|t|^{\frac{\alpha+\beta}{n+1}-1}, \quad u \neq 0, v \neq 0, t \neq 0. \end{aligned} \quad (3. 2)$$

Let $\mathbf{I}_{\alpha\beta\vartheta}f$ defined in (1. 7)-(1. 8) and

$$\frac{\alpha + \beta}{n + 1} = \frac{1}{p} - \frac{1}{q}, \quad 1 < p < q < \infty. \quad (3. 3)$$

By changing variable $\tau \longrightarrow \tau + \mu(u \cdot \eta - v \cdot \xi)$, we have

$$\begin{aligned} \mathbf{I}_{\alpha\beta\vartheta}f(u, v, t) &= \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \\ &\quad |u - \xi|^{\alpha-n} |v - \eta|^{\alpha-n} |t - \tau|^{\beta-1} \left[\frac{|u - \xi||v - \eta|}{|t - \tau|} + \frac{|t - \tau|}{|u - \xi||v - \eta|} \right]^{-\vartheta} d\xi d\eta d\tau \\ &\leq \iiint_{\mathbb{R}^{2n+1}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) \\ &\quad |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} |t - \tau|^{\frac{\alpha+\beta}{n+1}-1} d\xi d\eta d\tau \quad \text{by (3. 1)-(3. 2).} \end{aligned} \quad (3. 4)$$

Because $\mathbf{V}^{\alpha\beta\vartheta}$ is positive definite, it is suffice to assert $f \geq 0$. Define

$$\mathbf{F}_{\alpha\beta}(\xi, \eta, u, v, t) = \int_{\mathbb{R}} f(\xi, \eta, \tau - \mu(u \cdot \eta - v \cdot \xi)) |t - \tau|^{\frac{\alpha+\beta}{n+1}-1} d\tau. \quad (3. 5)$$

From (3. 4)-(3. 5), we find

$$\mathbf{I}_{\alpha\beta\vartheta}f(u, v, t) \leq \iint_{\mathbb{R}^{2n}} |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \mathbf{F}_{\alpha\beta}(\xi, \eta, u, v, t) d\xi d\eta. \quad (3. 6)$$

Recall the **Hardy-Littlewood-Sobolev theorem** stated in the beginning of this paper. By applying (1. 2) with $\mathbf{a} = \frac{\alpha+\beta}{n+1}$ and $\mathbf{N} = 1$, we have

$$\begin{aligned} \left\{ \int_{\mathbb{R}} \mathbf{F}_{\alpha\beta}^q(\xi, \eta, u, v, t) dt \right\}^{\frac{1}{q}} &\leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}} [f(\xi, \eta, t - \mu(u \cdot \eta - v \cdot \xi))]^p dt \right\}^{\frac{1}{p}} \\ &= \mathfrak{B}_{p,q} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} \end{aligned} \quad (3. 7)$$

regardless of $(u, v) \in \mathbb{R}^n \times \mathbb{R}^n$.

On the other hand, by applying (1. 2) with $\mathbf{a} = n[\frac{\alpha+\beta}{n+1}]$ and $\mathbf{N} = n$, we find

$$\left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |u - \xi|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} d\xi \right\}^q du \right\}^{\frac{1}{q}} \leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}^n} \|f(u, \eta, \cdot)\|_{L^p(\mathbb{R})}^p du \right\}^{\frac{1}{p}} \quad (3. 8)$$

and

$$\left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |v - \eta|^{n[\frac{\alpha+\beta}{n+1}]-n} \|f(\xi, \eta, \cdot)\|_{L^p(\mathbb{R})} d\eta \right\}^q dv \right\}^{\frac{1}{q}} \leq \mathfrak{B}_{p,q} \left\{ \int_{\mathbb{R}^n} \|f(\xi, v, \cdot)\|_{L^p(\mathbb{R})}^p dv \right\}^{\frac{1}{p}}. \quad (3. 9)$$

From (3. 6), we have

$$\begin{aligned}
& \|\mathbf{I}_{\alpha\beta\vartheta} f\|_{\mathbf{L}^q(\mathbb{R}^{2n+1})} \\
& \leq \left\{ \iiint_{\mathbb{R}^{2n+1}} \left\{ \iint_{\mathbb{R}^{2n}} |u - \xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} |v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} \mathbf{F}_{\alpha\beta}^q(\xi, \eta, u, v, t) d\xi d\eta \right\}^q dudvdt \right\}^{\frac{1}{q}} \\
& \leq \left\{ \iint_{\mathbb{R}^{2n}} \left\{ \iint_{\mathbb{R}^{2n}} |u - \xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} |v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} \left\{ \int_{\mathbb{R}} \mathbf{F}_{\alpha\beta}^q(\xi, \eta, u, v, t) dt \right\}^{\frac{1}{q}} d\xi d\eta \right\}^q dudv \right\}^{\frac{1}{q}} \\
& \quad \text{by Minkowski integral inequality} \\
& \leq \mathfrak{B}_{p\ q} \left\{ \iint_{\mathbb{R}^{2n}} \left\{ \iint_{\mathbb{R}^{2n}} |u - \xi|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} |v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} \|f(\xi, \eta, \cdot)\|_{\mathbf{L}^p(\mathbb{R})} d\xi d\eta \right\}^q dudv \right\}^{\frac{1}{q}} \quad \text{by (3. 7)} \\
& \leq \mathfrak{B}_{p\ q} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} \|f(u, \eta, \cdot)\|_{\mathbf{L}^p(\mathbb{R})} d\eta \right\}^p du \right\}^{\frac{q}{p}} dv \right\}^{\frac{1}{q}} \quad \text{by (3. 8)} \\
& \leq \mathfrak{B}_{p\ q} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} \left\{ \int_{\mathbb{R}^n} |v - \eta|^{n\left[\frac{\alpha+\beta}{n+1}\right]-n} \|f(u, \eta, \cdot)\|_{\mathbf{L}^p(\mathbb{R})} d\eta \right\}^q dv \right\}^{\frac{p}{q}} du \right\}^{\frac{1}{p}} \\
& \quad \text{by Minkowski integral inequality} \\
& \leq \mathfrak{B}_{p\ q} \left\{ \iint_{\mathbb{R}^{2n}} \|f(u, v, \cdot)\|_{\mathbf{L}^p(\mathbb{R})}^p dudv \right\}^{\frac{1}{p}} \quad \text{by (3. 9)} \\
& = \mathfrak{B}_{p\ q} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}.
\end{aligned} \tag{3. 10}$$

4 Proof of Theorem Two

Recall $\mathbf{M}_\gamma$ defined in (1. 11) for $0 \leq \gamma < 1$. By taking $\xi \longrightarrow u - \xi$, $\eta \longrightarrow v - \eta$ and $\tau \longrightarrow t - \tau$, $\mathbf{M}_\gamma$ can be equivalently defined as

$$\mathbf{M}_\gamma f(u, v, t) = \sup_{\mathbf{R} \ni (u, v, t)} \mathbf{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau. \tag{4. 11}$$

Similarly, $\mathbf{M}_{\alpha\beta}$ defined in (1. 13) is equivalent to

$$\begin{aligned}
& \mathbf{M}_{\alpha\beta} f(u, v, t) = \\
& \sup_{\substack{\mathbf{R} \ni (u, v, t) \\ \mathbf{vol}\{\mathbf{Q}_3\} = \mathbf{vol}\{\mathbf{Q}_1\}^{\frac{1}{n}} \mathbf{vol}\{\mathbf{Q}_2\}^{\frac{1}{n}}}} \mathbf{vol}\{\mathbf{Q}_1\}^{\frac{\alpha}{n}-1} \mathbf{vol}\{\mathbf{Q}_2\}^{\frac{\alpha}{n}-1} \mathbf{vol}\{\mathbf{Q}_3\}^{\beta-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau
\end{aligned} \tag{4. 12}$$

where $\mathbf{R} = \mathbf{Q}_1 \times \mathbf{Q}_2 \times \mathbf{Q}_3 \subset \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}$. Moreover, $\mathbf{vol}\{\mathbf{Q}_3\} = \mathbf{vol}\{\mathbf{Q}_1\}^{\frac{1}{n}} \mathbf{vol}\{\mathbf{Q}_2\}^{\frac{1}{n}}$ implies $\mathbf{vol}\{\mathbf{R}\} = \mathbf{vol}\{\mathbf{Q}_1\}^{1+\frac{1}{n}} \mathbf{vol}\{\mathbf{Q}_2\}^{1+\frac{1}{n}}$.

From (4. 12), we find

$$\begin{aligned}
\mathbf{M}_{\alpha\beta} f(u, v, t) &= \sup_{\substack{\mathbf{R} \ni (u, v, t) \\ \text{vol}\{\mathbf{Q}_3\} = \text{vol}\{\mathbf{Q}_1\}^{\frac{1}{n}} \text{vol}\{\mathbf{Q}_2\}^{\frac{1}{n}}}} \text{vol}\{\mathbf{Q}_1\}^{\left[\frac{\alpha+\beta}{n+1}-1\right]\left(1+\frac{1}{n}\right)} \text{vol}\{\mathbf{Q}_2\}^{\left[\frac{\alpha+\beta}{n+1}-1\right]\left(1+\frac{1}{n}\right)} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \\
&= \sup_{\substack{\mathbf{R} \ni (u, v, t) \\ \text{vol}\{\mathbf{Q}_3\} = \text{vol}\{\mathbf{Q}_1\}^{\frac{1}{n}} \text{vol}\{\mathbf{Q}_2\}^{\frac{1}{n}}}} \text{vol}\{\mathbf{R}\}^{\frac{\alpha+\beta}{n+1}-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \\
&\leq \sup_{\mathbf{R} \ni (u, v, t)} \text{vol}\{\mathbf{R}\}^{\gamma-1} \iiint_{\mathbf{R}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \quad (\gamma = \frac{\alpha+\beta}{n+1}) \\
&= \mathbf{M}_{\gamma} f(u, v, t).
\end{aligned} \tag{4. 13}$$

Hence, $\mathbf{M}_{\alpha\beta}$ is controlled by the strong fractional maximal operator $\mathbf{M}_{\gamma}$ whenever $\gamma = \frac{\alpha+\beta}{n+1}$. Let $\gamma = \frac{1}{p} - \frac{1}{q}$, $1 < p \leq q < \infty$. This required homogeneity condition can be found by changing dilation in (1. 12). In order to prove the converse, we need the following multi-parameter covering lemma.

Córdoba-Fefferman covering lemma

Let $\{\mathbf{R}_j\}_{j=1}$ be a collection of rectangles in $\mathbb{R}^{2n+1}$ parallel to the coordinates. There is a subsequence $\{\widehat{\mathbf{R}}_k\}_{k=1}$ such that

$$\text{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \lesssim \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \tag{4. 14}$$

and

$$\left\|\sum_k \chi_{\widehat{\mathbf{R}}_k}\right\|_{L^p(\mathbb{R}^{2n+1})}^p \lesssim \text{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}, \quad 1 < p < \infty \tag{4. 15}$$

where χ is an indicator function.

Remark 4.1. This covering lemma is established by Córdoba and Fefferman [3] within a much more general setting. Namely, the Lebesgue measure is replaced by an absolutely continuous measure satisfying the rectangle A_{∞} -property. In the following subsection, we prove this covering lemma w.r.t Lebesgue measure within a more direct approach.

Define

$$\mathbf{U}_{\lambda} = \left\{(u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_{\gamma} f(u, v, t) > \lambda\right\}. \tag{4. 16}$$

Given any $(u, v, t) \in \mathbf{U}_{\lambda}$, there is a rectangle $\mathbf{R}_j \ni (u, v, t)$ such that

$$\text{vol}\{\mathbf{R}_j\}^{\gamma-1} \iiint_{\mathbf{R}_j} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau > \frac{1}{2} \lambda. \tag{4. 17}$$

Let (u, v, t) run through the set $\mathbf{U}_{\lambda}$. We have $\mathbf{U}_{\lambda} \subset \bigcup_j \mathbf{R}_j$.

By applying the covering lemma, we select a subsequence $\{\widehat{\mathbf{R}}_k\}_{k=1}^\infty$ from the union above and

$$\begin{aligned}
\mathbf{vol}\left\{\mathbf{U}_\lambda\right\} &\lesssim \mathbf{vol}\left\{\bigcup_j \mathbf{R}_j\right\} \lesssim \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} \quad \text{by (4. 14)} \\
&\leq \sum_k \mathbf{vol}\left\{\widehat{\mathbf{R}}_k\right\} \\
&\leq \sum_k \left\{ \frac{2}{\lambda} \iiint_{\widehat{\mathbf{R}}_k} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \right\}^{\frac{1}{1-\gamma}} \quad \text{by (4. 17).}
\end{aligned} \tag{4. 18}$$

Because $0 \leq \gamma < 1$, we further have

$$\begin{aligned}
\mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\} &\lesssim \lambda^{-\frac{1}{1-\gamma}} \left\{ \sum_k \iiint_{\widehat{\mathbf{R}}_k} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))| d\xi d\eta d\tau \right\}^{\frac{1}{1-\gamma}} \\
&= \lambda^{-\frac{1}{1-\gamma}} \left\{ \iiint_{\mathbb{R}^{2n+1}} \left| f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi)) \sum_k \chi_{\widehat{\mathbf{R}}_k}(\xi, \eta, \tau) \right| d\xi d\eta d\tau \right\}^{\frac{1}{1-\gamma}} \\
&\leq \lambda^{-\frac{1}{1-\gamma}} \left\{ \iiint_{\mathbb{R}^{2n+1}} |f(\xi, \eta, \tau + \mu(u \cdot \eta - v \cdot \xi))|^p d\xi d\eta d\tau \right\}^{\frac{1}{p} \frac{1}{1-\gamma}} \left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}^{\frac{1}{1-\gamma}} \\
&\quad \text{by Hölder inequality} \\
&= \lambda^{-\frac{1}{1-\gamma}} \left\{ \iint_{\mathbb{R}^{2n}} \|f(\xi, \eta, \cdot)\|_{\mathbf{L}^p(\mathbb{R})}^p d\xi d\eta \right\}^{\frac{1}{p} \frac{1}{1-\gamma}} \left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})}^{\frac{1}{1-\gamma}} \\
&\leq \lambda^{-\frac{1}{1-\gamma}} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}^{\frac{1}{1-\gamma}} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{p-1}{p} \frac{1}{1-\gamma}} \quad \text{by (4. 15).}
\end{aligned} \tag{4. 19}$$

By raising both sides of (4. 19) to the $(1 - \gamma)$ -th power and then taking into account for $1 - \gamma - \frac{p-1}{p} = \frac{1}{p} - \left[\frac{1}{p} - \frac{1}{q}\right] = \frac{1}{q}$, we find

$$\mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{q}} \lesssim \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}. \tag{4. 20}$$

Let $\mathbf{U}_\lambda$ defined in (4. 16). From (4. 18) and (4. 20), we obtain

$$\begin{aligned}
\mathbf{vol}\left\{(u, v, t) \in \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R} : \mathbf{M}_\gamma f(u, v, t) > \lambda\right\}^{\frac{1}{q}} &\lesssim \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{q}} \\
&\lesssim \frac{1}{\lambda} \|f\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})}.
\end{aligned} \tag{4. 21}$$

By using this weak type (p, q) -estimate and applying Marcinkiewicz interpolation theorem, we conclude that $\mathbf{M}_\gamma$ is bounded from $\mathbf{L}^p(\mathbb{R}^{2n+1})$ to $\mathbf{L}^q(\mathbb{R}^{2n+1})$ if $\gamma = \frac{1}{p} - \frac{1}{q}$, $1 < p \leq q < \infty$.

4.1 Proof of the covering lemma

We select $\widehat{\mathbf{R}}_k$ from $\{\mathbf{R}_j\}_{j=1}$ as follows.

Let $\widehat{\mathbf{R}}_1 = \mathbf{R}_1$. Having chosen $\widehat{\mathbf{R}}_1, \widehat{\mathbf{R}}_2, \dots, \widehat{\mathbf{R}}_{N-1}$, we pick $\widehat{\mathbf{R}}_N$ as the first rectangle $\mathbf{R}$ on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_{N-1}$ so that

$$\mathbf{vol} \left\{ \mathbf{R} \cap \bigcup_{k=1}^{N-1} \widehat{\mathbf{R}}_k \right\} \leq \frac{1}{2} \mathbf{vol} \{\mathbf{R}\}. \quad (4.22)$$

Suppose $\mathbf{R}$ is an unselected rectangle. There is a positive number M such that $\mathbf{R}$ is on the list of $\mathbf{R}_j$'s after $\widehat{\mathbf{R}}_M$ and

$$\mathbf{vol} \left\{ \mathbf{R} \cap \bigcup_{k=1}^M \widehat{\mathbf{R}}_k \right\} > \frac{1}{2} \mathbf{vol} \{\mathbf{R}\}. \quad (4.23)$$

Let $\mathbf{M}$ be the strong maximal operator defined in $\mathbb{R}^{2n+1}$. Observe that (4.23) further implies

$$\mathbf{M} \chi_{\bigcup_k \widehat{\mathbf{R}}_k}(u, v, t) > \frac{1}{2}, \quad (u, v, t) \in \bigcup_{j=1} \mathbf{R}_j. \quad (4.24)$$

Indeed, if $\mathbf{R}_j$ is selected as $\widehat{\mathbf{R}}_k$ for some k , then $\mathbf{M} \chi_{\bigcup_k \widehat{\mathbf{R}}_k}(u, v, t) \geq 1$ whenever $(u, v, t) \in \mathbf{R}_j$.

From (4.24), by applying the L^p -boundedness of $\mathbf{M}$, we have

$$\begin{aligned} \mathbf{vol} \left\{ \bigcup_j \mathbf{R}_j \right\} &= \iiint_{\bigcup_j \mathbf{R}_j} dudvdt \\ &\leq 2^2 \iiint_{\bigcup_j \mathbf{R}_j} [\mathbf{M} \chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \\ &\lesssim \iiint_{\mathbb{R}^{2n+1}} [\mathbf{M} \chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \\ &\lesssim \int_{\mathbb{R}^{2n+1}} [\chi_{\bigcup_k \widehat{\mathbf{R}}_k}]^2(u, v, t) dudvdt \\ &= \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} dudvdt = \mathbf{vol} \left\{ \bigcup_k \widehat{\mathbf{R}}_k \right\}. \end{aligned} \quad (4.25)$$

On the other hand, (4.22) implies

$$\mathbf{vol} \left\{ \widehat{\mathbf{R}}_N \cap \bigcup_{k=1}^{N-1} \widehat{\mathbf{R}}_k \right\} \leq \frac{1}{2} \mathbf{vol} \{\widehat{\mathbf{R}}_N\}. \quad (4.26)$$

Denote $\widehat{\mathbf{E}}_N = \widehat{\mathbf{R}}_N \setminus \bigcup_{k=1}^{N-1} \widehat{\mathbf{R}}_k$ for every $N \geq 1$. From (4.26), we find

$$\mathbf{vol} \{\widehat{\mathbf{E}}_N\} > \frac{1}{2} \mathbf{vol} \{\widehat{\mathbf{R}}_N\}. \quad (4.27)$$

Let $\phi \in \mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})$ and $\|\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} = 1$. We have

$$\begin{aligned}
\iiint_{\mathbb{R}^{2n+1}} \phi(u, v, t) \sum_k \chi_{\widehat{\mathbf{R}}_k}(u, v, t) dudvdt &= \sum_k \iiint_{\widehat{\mathbf{R}}_k} \phi(u, v, t) dudvdt \\
&= \sum_k \left\{ \frac{1}{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}} \iiint_{\widehat{\mathbf{R}}_k} \phi(u, v, t) dudvdt \right\} \mathbf{vol}\{\widehat{\mathbf{R}}_k\} \\
&< 2 \sum_k \left\{ \frac{1}{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}} \iiint_{\widehat{\mathbf{R}}_k} \phi(u, v, t) dudvdt \right\} \mathbf{vol}\{\widehat{\mathbf{E}}_k\} \quad \text{by (4. 27)} \\
&\lesssim \sum_k \iiint_{\widehat{\mathbf{E}}_k} \left\{ \frac{1}{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}} \int_{\widehat{\mathbf{R}}_k} \phi(u, v, t) dudvdt \right\} d\xi d\eta d\tau \\
&\lesssim \sum_k \iiint_{\widehat{\mathbf{E}}_k} \mathbf{M}\phi(\xi, \eta, \tau) d\xi d\eta d\tau \\
&= \iiint_{\bigcup_k \widehat{\mathbf{E}}_k} \mathbf{M}\phi(\xi, \eta, \tau) d\xi d\eta d\tau \\
&= \iiint_{\bigcup_k \widehat{\mathbf{R}}_k} \mathbf{M}\phi(\xi, \eta, \tau) d\xi d\eta d\tau.
\end{aligned} \tag{4. 28}$$

Observe that

$$\frac{1}{\mathbf{vol}\{\widehat{\mathbf{R}}_k\}} \iiint_{\widehat{\mathbf{R}}_k} \phi(u, v, t) dudvdt \leq \mathbf{M}\phi(\xi, \eta, \tau) \quad \text{whenever } (\xi, \eta, \tau) \in \widehat{\mathbf{R}}_k. \tag{4. 29}$$

By applying Hölder inequality and using the $\mathbf{L}^p$ -boundedness of $\mathbf{M}$, we find

$$\begin{aligned}
\iiint_{\bigcup_k \widehat{\mathbf{R}}_k} \mathbf{M}_\mu \phi(\xi, \eta, \tau) d\xi d\eta d\tau &\leq \|\mathbf{M}\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}} \\
&\leq \mathfrak{B}_p \|\phi\|_{\mathbf{L}^{\frac{p}{p-1}}(\mathbb{R}^{2n+1})} \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}} \\
&= \mathfrak{B}_p \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}.
\end{aligned} \tag{4. 30}$$

By substituting (4. 30) to (4. 28) and taking the supremum of ϕ , we arrive at

$$\left\| \sum_k \chi_{\widehat{\mathbf{R}}_k} \right\|_{\mathbf{L}^p(\mathbb{R}^{2n+1})} \leq \mathfrak{B}_p \mathbf{vol}\left\{\bigcup_k \widehat{\mathbf{R}}_k\right\}^{\frac{1}{p}}. \tag{4. 31}$$

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